Will Ruff

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Education

San Diego State University, Department of Computer Science
Bachelor of Science in Computer Science
Cumulative GPA: 3.83
Expected Graduation: May 2026
Relevant Coursework: Advanced Programming Languages • Operating Systems • Computer Architecture • Data Structures • Intro to Software Systems • Computer Organization • Discrete Mathematics • Linear Algebra
Skills: Python, C++, Java, GoLang, HTML5, JavaScript, CSS, GitLab CI/CD, Docker, Data Preprocessing, Shell Scripting, Object-Oriented Programming, Algorithm Design and Analysis, Version Control (Git), API Integration, Computer Architecture, Debugging and Testing, Data Analytics, Software Development Lifecycle (SDLC)

Work Experience

Glympse — Seattle, Washington
Location Sharing Technology
Software Engineering Intern — June 2024 – Present

- Developed Python and Golang scripts for file comparison, cutting memory usage by 25% via benchmarking.
- Migrated legacy code to a new repository, adding essential parameters and enhancing report functionality.
- Refactored and modernized a shell script report in Python, enhancing readability and maintainability.
- Built and implemented comprehensive unit tests to ensure the reliability and accuracy of the Python code.
- Deployed code using GitLab CI/CD pipelines, configuring environments with '.gitlab-ci.yml' and Docker.

San Diego State University — San Diego, California
Academic Institution
Student Research Assistant — February 2024 – June 2024

- Conducted extensive research under faculty in the Integrating Cyber Innovations with Physical World Lab.
- Investigated detailed medical device recalls to detect underlying problems related to software issues.
- Implemented efficient code to automate data extraction from the FDA recall website for relevant analysis.
- Engaged in collaborative problem-solving exercises to address research challenges and advance objectives.
- Utilized HTML5 to design a compelling and operational website that proficiently showcases research.

Treobytes — San Diego, California
STEM Education Program
Facilitator — March 2024 – May 2024

- Developed and customized innovative STEM lesson plans to suit the interests of diverse student groups.
- Facilitated intriguing educational sessions for students, delivering course content effectively and engagingly.
- Conducted meticulous and thorough quality assessments on STEM resources for accuracy and functionality.
- Demonstrated exceptional organizational and time management skills to balance multiple responsibilities.
- Assisted in the development of a coding course where students learned by creating their own video games.

Projects

StockScript — November 2024 – December 2024

- Designed and implemented a domain-specific programming language for stock and crypto modeling.
- Developed grammar definitions using textX and created a Python interpreter to execute scripts effectively.
- Created user-friendly stock and crypto declarations with operations like price changes, merges, and splits.
- Developed display features to present asset data, including prices, market caps, and supply metrics.
- Added a simulation mode to generate good, bad, or random days for specific stocks or the entire market.

Restaurant Flow Control — November 2024

- Designed a restaurant simulation to manage seating and customer flow with a producer-consumer model.
- Utilized a buffer to coordinate threads for seat request types, ensuring efficient resource allocation.
- Used mutexes and semaphores to ensure thread safety and prevent deadlocks in multithreaded operations.
- Critically analyzed resource utilization metrics to identify bottlenecks and improve overall system efficiency.
- Implemented a configurable CLI for testing various scenarios and analyzing the overall system performance.

Spam Detection AI Model — September 2023 – December 2023

- Collaborated with a team of five individuals to develop an AI Model designed to detect spam messages.
- Implemented a training set comprised of over 10,000 messages to effectively train our model using Python.
- Preprocessed the dataset thoroughly and implemented an algorithm based on Naive Bayes Classification.
- Achieved a remarkable 98% accuracy in detecting spam messages through rigorous debugging and testing.
- Created an engaging slide deck and confidently presented the final project to the Artificial Intelligence Club.

Additional Information

Honors: Dean's List (3/4 Semesters), High School Valedictorian & Faculty Selected Master of Ceremonies (June '22), 1st Place DECA International Career Development (Business Law & Ethics Roleplay)(May '21)
Organizations: Artificial Intelligence Club, ELeet Code Club, Undergraduate Student Research, Pickleball Club
Interests: Entrepreneurship, Lifting Weights, Pickleball, Basketball, Cooking, Chess, Camping, Running